

Physics-Based Modeling of Interfacial Polar-Phase Nucleation and Piezoelectric Coupling in PVDF–Metal Oxide Nanocomposites for Energy Harvesting

[Author names and affiliations]

Abstract

We develop a first-order, physics-based framework for analysing PVDF/metal-oxide nanocomposites for piezoelectric energy harvesting. The framework links PVDF phase content, interphase geometry, effective dielectric response, 0–3 piezoelectric coupling, and lossy electrical output under matched resistive loading. All results are model predictions based on explicitly stated assumptions and have not yet been experimentally validated. They are benchmarked against published measurements where these exist.

Within the parameter range considered, the model indicates that nanoparticle-induced changes in PVDF polar-phase content can dominate the direct piezoelectric contribution of high-permittivity inclusions. For bulk-like BaTiO₃ in PVDF, the predicted dilute-limit local-field coefficient is about 0.021. For BaTiO₃ nanocrystals, whose measured permittivity is only about 80–300, it rises to 0.16–0.35. The dominance of polar-phase content then holds only if the filler d_{33} falls in proportion to its permittivity, under the adopted electrostrictive scaling assumption.

Results are reported across the assumed nucleation scenarios. At 10 vol% ZnO, the predicted $d_{33}g_{33}$ varies from 2.13 to 7.68×10^{-12} m²/N, compared with 2.55×10^{-12} m²/N for unfilled PVDF. Weighting the γ -phase below β can remove the gain entirely.

Morris global screening identifies matrix d_{33} , poling efficiency and maximum polar-phase gain as the dominant parameters, with strong interactions. Particle-size calculations predict monotonic improvement for PVDF/ZnO toward the nanoscale. For the assumed interphase permittivity $\varepsilon_i = 30$, PVDF/BaTiO₃ instead has a model-predicted optimum near 17 nm. A fit to published phase data for one electrospun PVDF/ZnO system supports a saturating nucleation law, but saturation occurs below 1 vol%. Letting crystallinity vary as measured in that system raises the predicted d_{33} gain at 1 vol% from 1.2 to 1.8–2.0 times the unfilled film. The framework is thus checked at the level of individual assumptions, with full experimental validation proposed. It provides a quantitative basis for designing and experimentally validating PVDF–oxide nanocomposite harvesters.

Keywords: PVDF; β -phase; nanocomposite; interphase; local field; Maxwell–Wagner–Sillars polarization; piezoelectric energy harvesting

1 Introduction

Wearable sensors, implants and distributed sensor nodes need microwatt- to milliwatt-level power without battery replacement. Mechanical energy from body motion, vibration and flow is widely available at 1–100 Hz, and piezoelectric nanogenerators convert it directly into electrical charge [17].

PVDF is attractive for this role. Its $-\text{CH}_2-\text{CF}_2-$ repeat unit carries a dipole of about 2.1 D, and the polymer is flexible, chemically stable and easy to process [1, 3]. Its piezoelectric coefficient is modest ($|d_{33}| \approx 20\text{--}35$ pC/N), but its low permittivity ($\varepsilon_r \approx 10\text{--}12$) gives a voltage coefficient $|g_{33}|$ of 0.2–0.33 V m/N. The piezoelectric response of a PVDF film is limited by its crystallinity X_c , by the fraction of crystals in a polar phase, and by how completely those crystals

are poled. Melt-processed PVDF crystallizes mostly in the non-polar α -phase, and metal-oxide nanoparticles are widely added to raise the polar fraction [3].

Several mechanisms are often conflated when such composites are evaluated: polar-phase nucleation at the interface, the filler’s own piezoelectricity, interfacial space charge, and triboelectric artefacts in device tests. In addition, the FTIR electroactive fraction is frequently reported as the β -fraction although the 840 cm^{-1} band includes the γ -phase [4].

This paper does not report new measurements. Its contribution is a modeling framework, with five elements:

1. a single chain linking phase content, interphase geometry, dielectric mixing, 0–3 piezoelectric coupling and lossy device output;
2. explicit treatment of the sign of d_{33} and of the filler poling direction;
3. sensitivity analyses over the nucleation law, particle size, interphase properties, poling efficiency, dielectric loss and frequency, which show where the predictions are robust and where they are not;
4. global (Morris) screening and a paired Latin-hypercube comparison that account for parameter interactions;
5. a benchmark of individual assumptions against published measurements, and an open, tested implementation with a script that regenerates every figure and quoted number and a fitting routine that replaces the assumed nucleation law with measured data.

2 Polymorphism and phase quantification

Only the polar β - and γ -phases contribute to piezoelectricity, so the harvesting-relevant quantity is the polar crystalline content $X_c F_{\text{polar}}$. Table 1 lists the relevant phases [1, 3, 4].

Table 1: Main PVDF polymorphs and their signatures (XRD for Cu $K\alpha$).

Phase	Conformation	Polarity	FTIR bands (cm^{-1})	XRD 2θ ($^\circ$)
α (II)	TGTG'	non-polar	614, 763, 795, 976	17.7, 18.4, 19.9, 26.6
β (I)	all-trans	strongly polar	840, 1275 (β only)	20.6–20.8
γ (III)	T ₃ GT ₃ G'	moderately polar	833–840, 1234 (γ only)	18.5, 19.2, 20.0

With the Lambert–Beer law, the electroactive fraction of the crystalline phase follows from the absorbances at 763 and 840 cm^{-1} [2]:

$$F_{EA} = \frac{A_{840}}{(K_{840}/K_{763})A_{763} + A_{840}}, \quad K_{763} = 6.1 \times 10^4, \quad K_{840} = 7.7 \times 10^4 \text{ cm}^2 \text{ mol}^{-1}. \quad (1)$$

Following Cai *et al.* [4], F_{EA} is split using the peak-to-valley heights ΔH of the β -only 1275 cm^{-1} and γ -only 1234 cm^{-1} bands:

$$F(\beta) = F_{EA} \frac{\Delta H_{1275}}{\Delta H_{1275} + \Delta H_{1234}}, \quad F(\gamma) = F_{EA} \frac{\Delta H_{1234}}{\Delta H_{1275} + \Delta H_{1234}}. \quad (2)$$

The matrix crystallinity is normalised to polymer mass, with filler weight fraction w_f and cold-crystallization enthalpy ΔH_{cc} :

$$X_c = \frac{\Delta H_m - \Delta H_{cc}}{(1 - w_f) \Delta H_{100}}, \quad \Delta H_{100} \approx 104.7 \text{ J g}^{-1}. \quad (3)$$

FTIR-derived fractions depend on spectral preprocessing. Eq. (1) uses baseline-corrected absorbance *heights* at fixed wavenumbers, not integrated areas. The β/γ split uses peak-to-valley heights. Both require a consistent baseline across samples and the same normalization (film thickness or a reference band). When the 763 or 840 cm^{-1} bands overlap neighbouring features, they should be deconvoluted. The **phases** module takes preprocessed absorbances as inputs. The preprocessing protocol should be reported together with the fractions.

Because the γ -phase has a smaller remanent polarization than β , the polar fraction entering the piezoelectric model is weighted:

$$F_{\text{eff}} = F(\beta) + \alpha_\gamma F(\gamma), \quad 0 \leq \alpha_\gamma \leq 1. \quad (4)$$

No reliable value of α_γ has been established for PVDF composites, so α_γ and the γ share of the polar phase are treated as uncertain parameters (Sections 7.4 and 7.9). The baseline assumes that the nucleated polar phase is β , as reported for ferrite- and ZnO-nucleated PVDF [7, 16].

3 Interfacial physics

3.1 Interphase volume

We represent the interphase as a shell of thickness t around each sphere of radius r . Here t is a model parameter, not a measured thickness. At filler volume fraction ϕ , a random-overlap (Poisson) correction keeps the interphase within the available matrix volume:

$$\phi_i = (1 - \phi) \left[1 - \exp\left(-\frac{\phi [(1 + t/r)^3 - 1]}{1 - \phi}\right) \right], \quad (5)$$

which reduces to $\phi[(1 + t/r)^3 - 1]$ in the dilute limit. Throughout the paper, “particle size” means the diameter $d = 2r$. On a simple cubic lattice the mean surface-to-surface spacing is $s = d[(\pi/6\phi)^{1/3} - 1]$.

3.2 Interphase structure

Tanaka’s multi-core model describes the interphase as bonded (~ 1 nm), bound (2–9 nm) and loose (tens of nm) layers, overlapped by a diffuse electrical double layer [5]. Lewis argued that interfaces dominate dielectric behaviour at nanometric dimensions [6]. These are conceptual models of interaction zones, not universal fixed-thickness shells. In this paper, therefore, t and the interphase permittivity ε_i are treated as parameters and varied in Section 7.4.

In an idealised Debye–Hückel picture the extent of the diffuse layer is

$$\lambda_D = \sqrt{\frac{\varepsilon_r \varepsilon_0 k_B T}{2n z^2 e^2}}, \quad (6)$$

where n is the number density of mobile ions of valence z . This expression assumes mobile, thermally equilibrated carriers. In PVDF, charge is distributed among mobile ions, trapped space charge, carriers injected at electrodes, Maxwell–Wagner charge at internal boundaries, and fixed surface charge on the particles. Only the first of these is screened as above. We therefore use λ_D only as an order-of-magnitude indicator of whether double layers can overlap, not as a predictive length.

3.3 Mechanisms of polar-phase nucleation

Several mechanisms have been proposed for the stabilisation of β and γ conformations near filler surfaces.

Surface electrostatic interaction. For ferrite nanoparticles, Martins *et al.* attributed β -nucleation to the electrostatic interaction between particles with a negative zeta potential and the positively charged CH₂ groups [7]. Sebastian *et al.* found that positively charged surfaces can instead nucleate the β -phase through interaction with CF₂ dipoles, and that particle shape matters [8]. Either way, dipoles are pinned normal to the surface, which only all-trans or T₃G sequences accommodate. The strength of this bias can be estimated from the field $E = \sigma/(\epsilon_r \epsilon_0)$ outside a surface of charge density σ :

$$x = \frac{\mu E}{k_B T} = \frac{\mu \sigma}{\epsilon_r \epsilon_0 k_B T}, \quad \langle \cos \theta \rangle = \coth x - \frac{1}{x}. \quad (7)$$

For $\sigma = 0.01$ C/m², $\epsilon_r = 10$, $\mu = 2.1$ D and $T = 298$ K, $x \approx 0.19$ per monomer. Segments respond cooperatively over several monomers, so the bias per segment is larger. This is an estimate of plausibility, not a quantitative nucleation model.

Hydrogen bonding between surface hydroxyls and CF₂ groups, *confinement* of lamellar growth in narrow interparticle gaps, and *heterogeneous nucleation* that raises the crystallization temperature are also commonly invoked. *Epitaxial* crystallization has been proposed mainly for layered silicates. Lattice matching between filler and polymer is more complex than a comparison of spacings, so we do not use it quantitatively.

None of these mechanisms is modelled from first principles here. Their combined effect enters through a phenomenological nucleation law (Section 5.4).

4 Dielectric response

4.1 Effective-medium models

For dilute, isolated spheres the Maxwell–Garnett (MG) expression is

$$\epsilon_c = \epsilon_m \frac{\epsilon_f + 2\epsilon_m + 2\phi(\epsilon_f - \epsilon_m)}{\epsilon_f + 2\epsilon_m - \phi(\epsilon_f - \epsilon_m)}. \quad (8)$$

We compare it with the symmetric Bruggeman model, the Lichtenecker logarithmic rule and the Yamada model [9]. To include an interphase, each particle is replaced by a coated sphere with core fraction $q = [r/(r+t)]^3$:

$$\epsilon_{eq} = \epsilon_i \frac{(\epsilon_f + 2\epsilon_i) + 2q(\epsilon_f - \epsilon_i)}{(\epsilon_f + 2\epsilon_i) - q(\epsilon_f - \epsilon_i)}, \quad (9)$$

and the coated spheres are mixed by MG at volume fraction $\phi(1+t/r)^3$. Calculations are stopped, not extrapolated, when this coated volume fraction exceeds 0.5. Beyond that point the shells overlap extensively and isolated-inclusion mixing is not meaningful. At 5 vol% loading with $t = 5$ nm, the criterion requires $d \geq 8.7$ nm. The particle-size study therefore starts at 10 nm.

MG assumes dilute, non-interacting, well-dispersed spheres. At 20–30 vol% the following all become significant: particle–particle field interaction, agglomeration, connectivity and percolation, non-spherical shape, interphase overlap, and anisotropy. *Predictions above approximately 15 vol% should therefore be regarded as qualitative model extrapolations*, and they are shaded as such in the figures.

4.2 Loss and Maxwell–Wagner–Sillars polarization

Each phase is described by a complex permittivity $\epsilon^*(\omega) = \epsilon'(\omega) - j\epsilon''(\omega)$, with $\tan \delta = \epsilon''/\epsilon'$. DC conduction contributes $\epsilon'' = \sigma/(\omega\epsilon_0)$, so

$$\tan \delta_\sigma = \frac{\sigma}{\omega\epsilon_0\epsilon'}, \quad (10)$$

which grows as $1/f$. Substituting complex permittivities into Eq. (8) gives a Debye-type interfacial relaxation with time constant

$$\tau_{\text{MWS}} = \varepsilon_0 \frac{\varepsilon_f + 2\varepsilon_m - \phi(\varepsilon_f - \varepsilon_m)}{\sigma_f + 2\sigma_m - \phi(\sigma_f - \sigma_m)}. \quad (11)$$

During poling, interfacial charge can stabilise oriented dipoles and add an electret-like contribution to measured d_{33} . In operation, it raises low-frequency ε' and ε'' . The effect of loss on harvested power is quantified in Section 7.6.

5 Piezoelectric coupling model

5.1 Constitutive relations and sign

In strain-charge form along the poling axis,

$$D_3 = d_{33}T_3 + \varepsilon_0 \varepsilon_{33}^T E_3, \quad S_3 = s_{33}^E T_3 + d_{33}E_3. \quad (12)$$

Poled PVDF has $d_{33} < 0$; its microscopic origin was analysed by Katsouras *et al.* [10]. Oxide ceramics and ZnO poled in the same direction have $d_{33} > 0$.

We distinguish the *intrinsic* piezoelectric coefficient, which arises from lattice and dipolar response, from the *effective measured* d_{33} obtained with a quasi-static meter. The measured value can also include electret-like polarization from trapped space charge, electrode effects and mechanical artefacts. The model below predicts an intrinsic effective coefficient only. Comparing it with measured values therefore requires the controls listed in Section 9.

5.2 Which field and which stress

Several electric fields must be distinguished: the macroscopic applied field E_0 , the average field in the matrix, the field inside a particle, the strongly non-uniform field at the interface, the field during poling, and the (much smaller) field generated during mechanical operation. The coefficient used below concerns only the *average field inside a dilute spherical particle* relative to the composite average field. It is relevant chiefly to how well the filler is poled and how its polarization couples to the electrodes. It does not describe field concentration at the interface or in narrow gaps between particles.

For a dilute sphere, Furukawa's local field and stress coefficients are [11]

$$L_E = \frac{3\varepsilon_c}{2\varepsilon_c + \varepsilon_f}, \quad L_T = \frac{5c_f}{3c_m + 2c_f}, \quad (13)$$

with c_m , c_f the elastic moduli. L_T assumes perfect bonding and linear elasticity. Interfacial debonding, voids, agglomeration, viscoelasticity of the matrix and frequency-dependent moduli all reduce the effective stress transfer, so L_T is an upper estimate.

5.3 First-order phenomenological mixture model

We write the model-predicted effective coefficient as

$$d_{33}^c = (1 - \phi) d_m + s \phi L_E L_T |d_f|, \quad d_m = d_{\text{ref}} \eta \frac{X_c F_{\text{eff}}}{X_{c,\text{ref}} F_{\text{polar,ref}}}, \quad (14)$$

where $s = +1$ (filler poled parallel to the matrix), -1 (antiparallel) or 0 (unpoled), and η is the matrix poling efficiency. The reference values $d_{\text{ref}} = -28$ pC/N, $X_{c,\text{ref}} = 0.50$ and $F_{\text{polar,ref}} = 0.85$ represent a typical poled, predominantly β -phase film. Their uncertainty is explored in Sections 7.4 and 7.9.

The filler coefficient d_f is not independent of the filler permittivity. For a perovskite with spontaneous polarization P_s and electrostrictive coefficient Q , the phenomenological relation $d_{33} = 2Q\varepsilon_0\varepsilon_{33}P_s$ holds. We adopt it as a *scaling assumption*, not as an established law for nanoscale BaTiO₃: a nanocrystal with reduced permittivity is assumed to have a proportionally reduced d_{33} at fixed P_s ; this is an upper estimate, because P_s also falls at small sizes [13]. Because $L_E \approx 3\varepsilon_c/\varepsilon_f$ for $\varepsilon_f \gg \varepsilon_c$, the product $L_E d_f$ is then nearly independent of ε_f . We report results both with d_f scaled in this way and with d_f held at the bulk value of 190 pC/N, which is an unrealistic upper bound for nanocrystals.

Eq. (14) is a first-order mixture rule, not a homogenization of the full piezoelectric tensor. Its assumptions are:

1. the matrix term is weighted by its volume fraction and scales linearly with polar crystalline content;
2. the filler term is additive and linear in ϕ , which holds only in the dilute limit;
3. field and stress coupling are independent, so their coefficients multiply;
4. filler polarization is fully aligned when poled, with no orientation distribution;
5. the interphase has no piezoelectric response of its own beyond its contribution to F_{polar} ;
6. particles are spherical, so only the longitudinal coefficient d_{33} is treated.

Shear (d_{15}) and transverse (d_{31}) coefficients, anisotropic particles, and the coupling between the effective compliance and piezoelectric tensors require a full Eshelby-type or numerical homogenization, which is outside the scope of this paper.

Ploss *et al.* poled the inclusions and matrix of lead titanate/P(VDF-TrFE) 0–3 composites independently. At 27 vol% filler, parallel or antiparallel poling produced composites with vanishing piezoelectric or vanishing pyroelectric activity [12]. That work concerns a different filler and copolymer. It supports the sign argument used here, but it is not a direct validation for PVDF/BaTiO₃ or PVDF/ZnO.

5.4 Nucleation law

Polar-phase content enters Eq. (14) through a phenomenological law. As a function of loading,

$$F_{\text{polar}}(\phi) = F_0 + \Delta F \left[1 - e^{-\phi/\phi_{\text{sat}}} \right]. \quad (15)$$

For the particle-size study, nucleation is tied instead to the interphase volume of Eq. (5):

$$F_{\text{polar}}(\phi_i) = F_0 + \Delta F \left[1 - e^{-\phi_i/\phi_{i,\text{sat}}} \right]. \quad (16)$$

Both laws are assumptions, and this is the main limitation of the framework. Because the predicted gains in d_{33} follow from the assumed gains in F_{polar} , the model cannot by itself show that nucleation drives the improvement. Section 7.4 therefore reports results across four scenarios (Table 2), and Section 9 describes how measured FTIR/DSC data replace the assumed law.

5.5 Figures of merit

For off-resonance harvesting, the transduction figure of merit is

$$g_{33} = \frac{d_{33}}{\varepsilon_0\varepsilon_r}, \quad \text{FoM} = d_{33}g_{33} = \frac{d_{33}^2}{\varepsilon_0\varepsilon_r}. \quad (17)$$

This lossless figure of merit is complemented by the lossy device model below.

Table 2: Nucleation scenarios used in the sensitivity analysis (Eq. 15).

Scenario	F_0	ΔF	ϕ_{sat}
None	0.50	0	–
Weak	0.50	0.10	0.03
Nominal	0.50	0.35	0.03
Strong	0.50	0.45	0.015

6 Device model and computational methods

Below mechanical resonance, the film is represented as a charge source in parallel with its capacitance $C_p = \varepsilon_0 \varepsilon_r A / t_f$ and a loss conductance $G = \omega C_p \tan \delta + 1/R_{\text{leak}}$. Under a sinusoidal force with charge amplitude $Q_0 = d_{33}F_0$, the average power into a load R is

$$P(R) = \frac{(\omega Q_0)^2 R}{2[(1 + GR)^2 + (\omega C_p R)^2]}, \quad (18)$$

with matched load and maximum power

$$R_{\text{opt}} = \frac{1}{\sqrt{G^2 + (\omega C_p)^2}}, \quad P_{\text{max}} = \frac{(\omega Q_0)^2}{4[\sqrt{G^2 + (\omega C_p)^2} + G]}, \quad (19)$$

which reduce to $R_{\text{opt}} = 1/(\omega C_p)$ and $P_{\text{max}} = \omega Q_0^2/(4C_p)$ for $G = 0$. This linear, off-resonance model omits mechanical resonance, contact mechanics, hysteresis, parasitic capacitance and rectifier losses.

Table 3 lists the parameters. They are representative values, not measurements on a specific sample. The bulk BaTiO₃ permittivity of 1700 is an upper value for ceramics. Measured permittivities of BaTiO₃ nanocrystals are about 80 for paraelectric particles of 100 nm or less, and 150–300 for ferroelectric particles of 200 nm or more [14]. Both cases are analysed. The reference device is a 2 cm × 2 cm, 60 μm film loaded in compression by a 50 N sinusoidal force at 5 Hz. Matrix crystallinity is fixed at $X_c = 0.50$, typical of solution-cast PVDF, so that the effect of polar-phase content is isolated.

Table 3: Material parameters (representative room-temperature values).

Material	ε_r	$ d_{33} $ (pC/N)	Young’s modulus (GPa)
β-PVDF (poled)	12	28 (negative sign)	2.5
ZnO	8.8	12	140
BaTiO ₃ (bulk; nanocrystal range)	1700; 80–1700	190; scaled	110

All models are implemented in the open-source Python package `pvd_f_nano` (version 0.4.0). Its modules cover phases, interface, nucleation, dielectric response, piezoelectric coupling and harvester output. A suite of 48 unit tests checks limiting cases, Wiener bounds, consistency between independent MWS expressions, recovery of known nucleation-law parameters by the fitting routine, Morris screening on functions with known sensitivities, and agreement of analytical and numerical optima of $P(R)$ with and without loss. The script `examples/reproduce_paper.py` regenerates every figure and writes every quoted number to `paper/results/`.

7 Results

All results in this section are model predictions under the assumptions above.

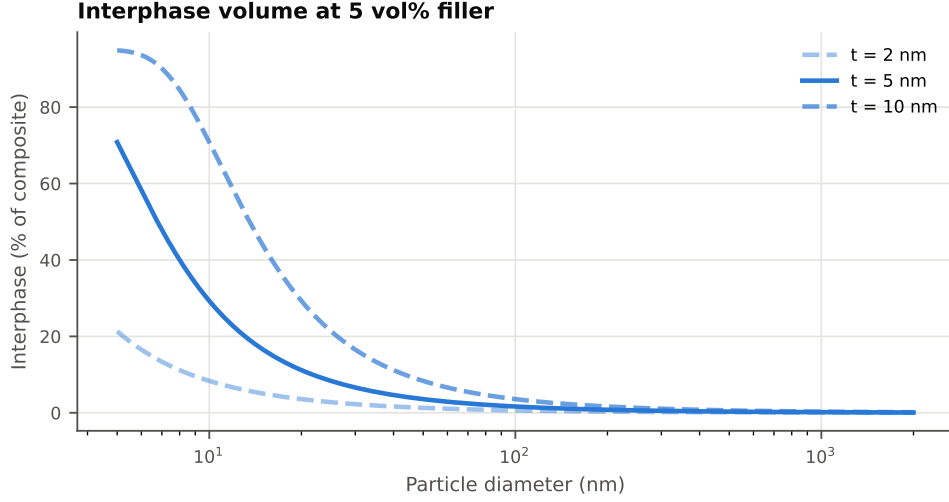


Figure 1: Model-predicted interphase volume fraction (Eq. 5) versus particle diameter at 5 vol% loading for assumed shell thicknesses of 2, 5 and 10 nm.

7.1 Interphase scaling

For a 5 nm shell at 5 vol%, the interphase occupies 11.2% of the composite for 20 nm particles, 29.3% for 10 nm particles and 0.15% for 1 μm particles (Fig. 1). Particles 20 nm in diameter are then on average 24 nm apart. These are geometric consequences of the assumed shell. Whether the shell is a physically distinct region must be established experimentally.

7.2 Permittivity and local field

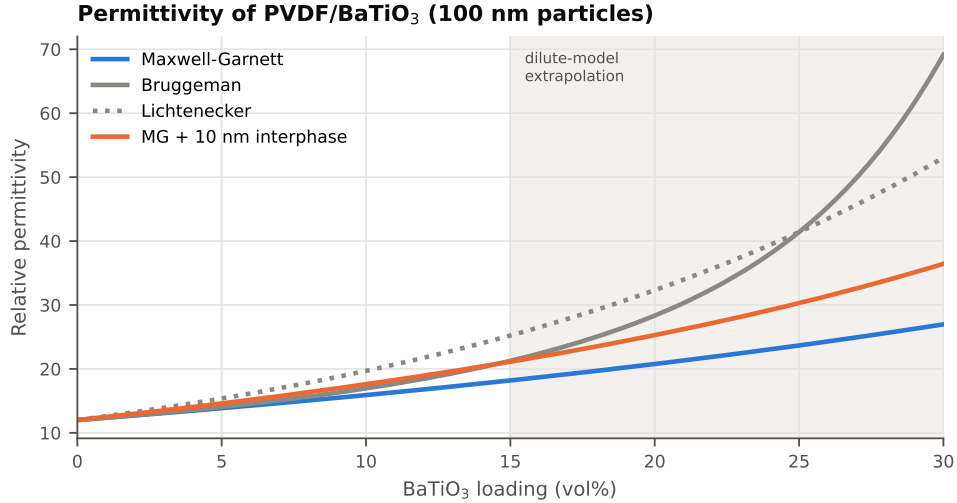


Figure 2: Model-predicted relative permittivity of PVDF/BaTiO₃ (100 nm particles). The shaded region marks the loading range where dilute-model predictions are extrapolations.

Up to 10 vol%, the four models agree within about 25%, and the predicted permittivity remains far below that of BaTiO₃ (Table 4, Fig. 2). Under the dilute-sphere model, a BaTiO₃ particle sees a small fraction of the average field: $L_E = 0.021$ in the dilute limit and 0.028 at 10 vol%, against 1.09 for ZnO. The stress coefficient is favourable for both ($L_T = 2.42$ for BaTiO₃, 2.44 for ZnO). Within this model, therefore, the direct contribution of an unpoled or

Table 4: Model-predicted relative permittivity of PVDF/BaTiO₃ (100 nm particles; interphase $t = 10$ nm, $\varepsilon_i = 30$).

BaTiO ₃ (vol%)	Maxwell–Garnett	Bruggeman	Lichtenecker	MG + interphase
5	13.9	14.1	15.4	14.6
10	15.9	17.0	19.7	17.6
20 [†]	20.8	28.3	32.3	25.3
30 [†]	27.0	69.1	53.0	36.4

[†]Qualitative extrapolation beyond the dilute range.

weakly poled high-permittivity filler is small. Real particles with non-spherical shapes, contacts or conductive pathways can experience substantially different local fields.

The small L_E depends on the bulk permittivity. For nanocrystal permittivities of 80 and 200, the dilute-limit L_E rises to 0.35 and 0.16. Table 5 shows the consequence at 10 vol% under the nominal nucleation scenario, where the matrix term is 24.8 pC/N. If the nanocrystals kept the bulk d_f of 190 pC/N, the filler term would reach 18.4 pC/N at $\varepsilon_f = 80$, comparable to the matrix term. With d_f scaled by permittivity, the filler term is 0.9–1.3 pC/N for all ε_f . The conclusion that polar-phase content dominates is therefore conditional on how d_f of the nanocrystals scales. Measuring it, for example by piezoresponse force microscopy on isolated particles, is a priority.

Table 5: Model-predicted effect of BaTiO₃ permittivity on the filler term at 10 vol% (nominal nucleation scenario; matrix term 24.8 pC/N).

ε_f	d_f rule	d_f (pC/N)	ε_c	L_E	Filler term (pC/N)	$ d_{33} $ antiparallel	$ d_{33} $ co-poled
80	bulk value	190	14.5	0.399	18.4	43.2	6.5
80	scaled with ε_f	8.9	14.5	0.399	0.9	25.7	24.0
200	bulk value	190	15.3	0.199	9.1	34.0	15.7
200	scaled with ε_f	22.4	15.3	0.199	1.1	25.9	23.8
1700	bulk value	190	15.9	0.028	1.3	26.1	23.6

7.3 Poling direction

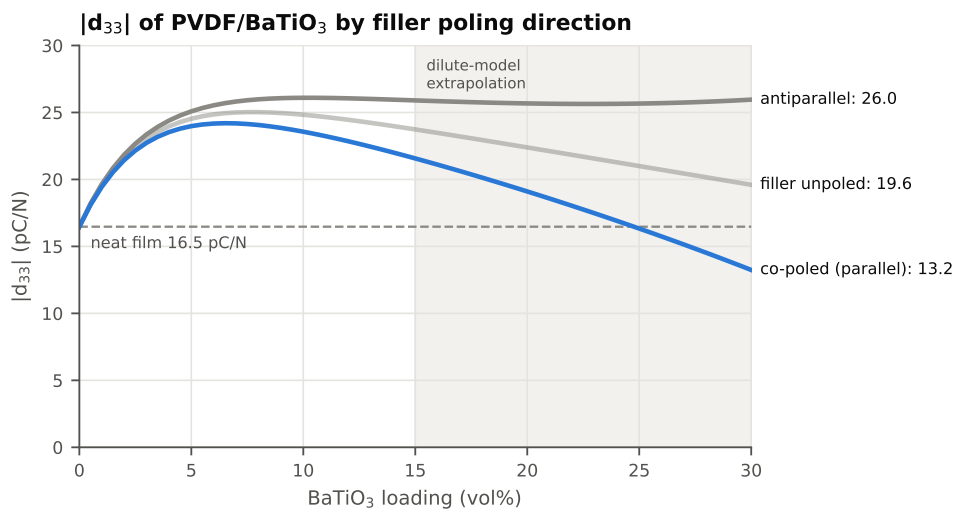


Figure 3: Model-predicted $|d_{33}|$ of PVDF/BaTiO₃ versus loading for three filler poling configurations (nominal nucleation scenario).

In the nominal nucleation scenario, all three poling cases rise from 16.5 to about 24–26 pC/N

by 6–8 vol% (Fig. 3). The rise comes from the assumed increase in F_{polar} . At 10 vol%, the predicted $|d_{33}|$ is 26.1 (antiparallel), 24.8 (unpoled filler) and 23.6 pC/N (co-poled). In the extrapolated range at 30 vol%, co-poling lowers $|d_{33}|$ to 13.2 pC/N, below the unfilled film.

7.4 Sensitivity to the nucleation law

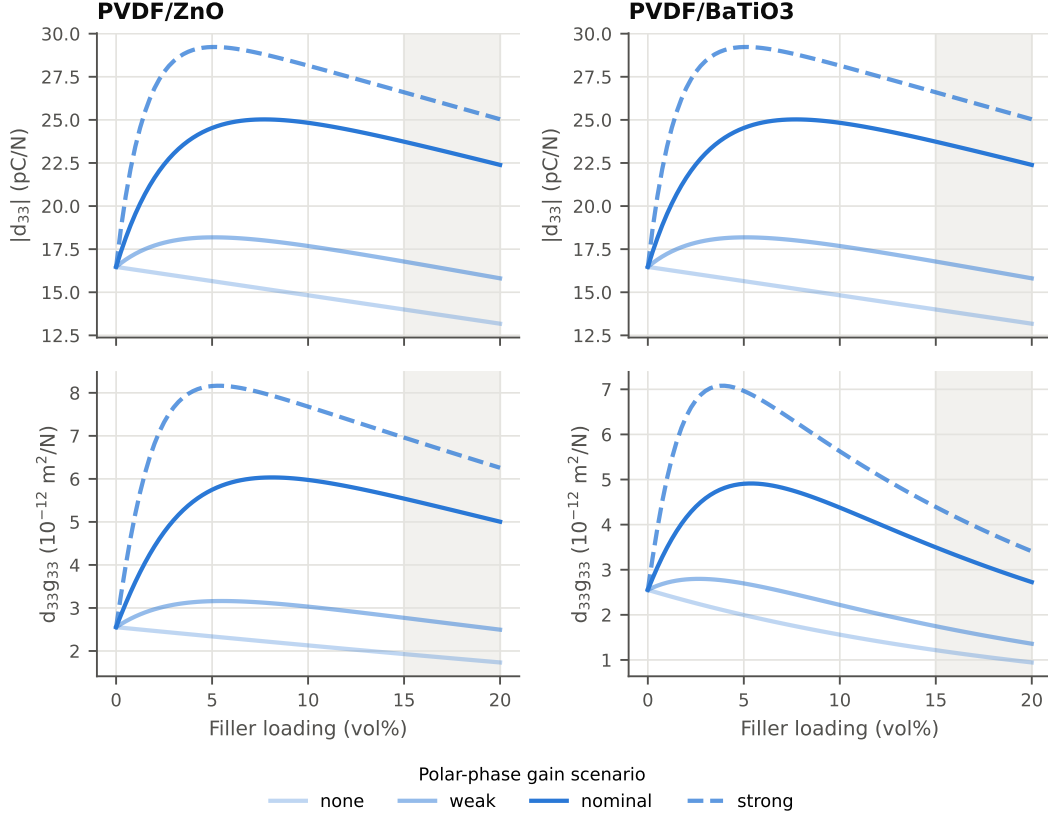


Figure 4: Model-predicted $|d_{33}|$ (top) and $d_{33}g_{33}$ (bottom) for PVDF/ZnO and PVDF/BaTiO₃ (filler unpoled) under the four nucleation scenarios of Table 2.

The nucleation law controls the predictions (Fig. 4, Table 7). Without a polar-phase gain, adding either filler lowers $|d_{33}|$, because it dilutes the piezoelectric matrix. Both fillers give the same $|d_{33}|$ in all scenarios, because the unpoled filler term is zero and the nucleation law is the same for both. Their figures of merit differ only through permittivity. For BaTiO₃, a weak polar-phase gain still gives a lower figure of merit than the unfilled film (2.22 vs $2.55 \times 10^{-12} \text{ m}^2/\text{N}$); for ZnO it gives a higher one (3.03).

The phase composition of the nucleated polar fraction matters as much as its size (Table 6). If half of it is γ -phase with $\alpha_\gamma = 0.25$, the nominal PVDF/ZnO figure of merit falls from 5.98 to $2.33 \times 10^{-12} \text{ m}^2/\text{N}$. That is below the unfilled film. Separating $F(\beta)$ from $F(\gamma)$ experimentally is therefore essential, not optional.

7.5 Particle size

When nucleation is tied to interphase volume, the predicted response depends strongly on particle size (Fig. 5). For PVDF/ZnO, the figure of merit rises from 2.44×10^{-12} at $1 \mu\text{m}$ to $6.74 \times 10^{-12} \text{ m}^2/\text{N}$ at 10 nm . Over the same range, P_{max} rises from 7.2 to 19.9 nW . For the assumed BaTiO₃ interphase permittivity $\varepsilon_i = 30$, the model predicts a figure-of-merit maximum of $4.59 \times 10^{-12} \text{ m}^2/\text{N}$ near 17 nm . This optimum is a consequence of the assumed interphase

Table 6: Model-predicted effect of γ -phase share and activity α_γ (Eq. 4) for PVDF/ZnO at 10 vol% (nominal scenario, filler unpoled). The γ share is irrelevant when $\alpha_\gamma = 1$.

γ share	α_γ	F_{eff}	$ d_{33} $ (pC/N)	$d_{33}g_{33}$ (10^{-12} m ² /N)
0	any	0.838	24.8	5.98
0.25	0.5	0.733	21.7	4.58
0.25	0.25	0.680	20.2	3.94
0.5	0.5	0.628	18.6	3.36
0.5	0.25	0.523	15.5	2.33

Table 7: Model-predicted response at 10 vol% loading (filler unpoled) for each nucleation scenario. Unfilled film: $|d_{33}| = 16.5$ pC/N, $d_{33}g_{33} = 2.55 \times 10^{-12}$ m²/N.

Scenario	$ d_{33} $ (pC/N)	$d_{33}g_{33}$, ZnO (10^{-12} m ² /N)	$d_{33}g_{33}$, BaTiO ₃ (10^{-12} m ² /N)
None	14.8	2.13	1.56
Weak	17.7	3.03	2.22
Nominal	24.8	5.98	4.38
Strong	28.1	7.68	5.63

5 vol% filler, 5 nm interphase, interphase-controlled nucleation

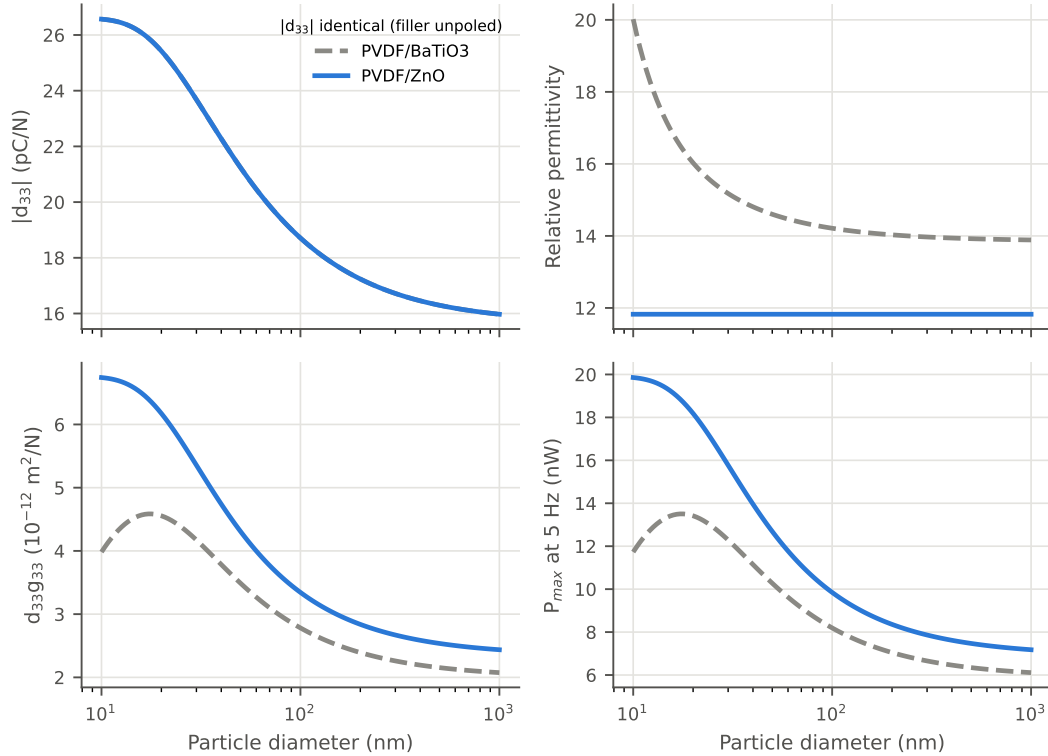


Figure 5: Model-predicted $|d_{33}|$, permittivity, $d_{33}g_{33}$ and P_{max} versus particle diameter at 5 vol%, with nucleation tied to interphase volume (Eq. 16; $t = 5$ nm, $\Delta F = 0.35$, $\phi_{i,\text{sat}} = 0.05$). Interphase permittivity: $\varepsilon_i = \varepsilon_m$ for ZnO, $\varepsilon_i = 30$ for BaTiO₃.

properties and nucleation law, not an experimentally optimized size. Below that size, the high-permittivity interphase raises the composite permittivity (to 20.0 at 10 nm) faster than nucleation raises d_{33} . Particles below 10 nm are not shown, because they violate the coated-sphere validity criterion of Section 4.1. With $\varepsilon_i = \varepsilon_m$ the BaTiO₃ curve would rise monotonically, like ZnO.

7.6 Frequency and dielectric loss

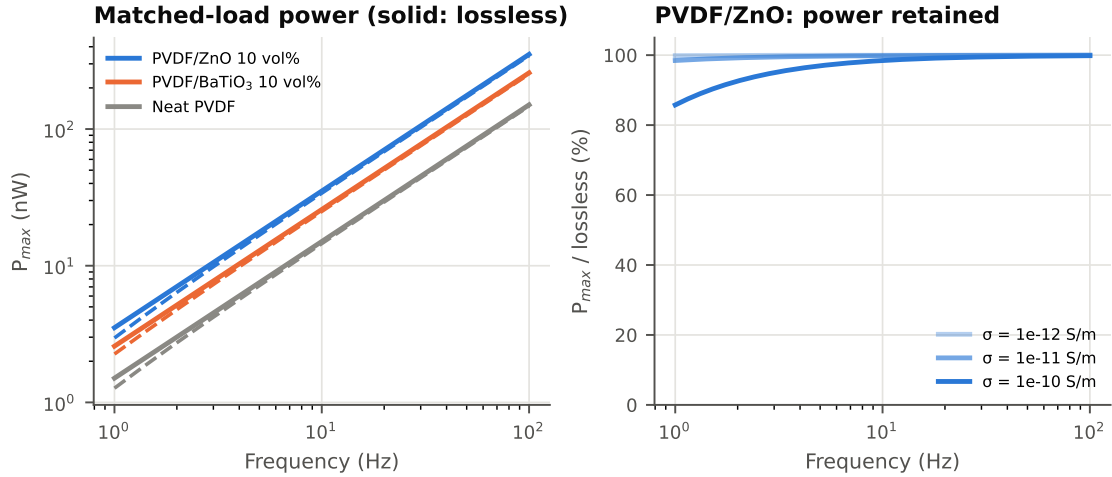


Figure 6: Left: model-predicted matched-load power from 1 to 100 Hz without loss (solid) and with $\sigma = 10^{-10}$ S/m plus $\tan \delta = 0.02$ (dashed). Right: fraction of lossless power retained by PVDF/ZnO (10 vol%) for three conductivities.

For the adopted off-resonance charge-source model under constant force amplitude, $P_{\max} \propto f$. Real devices can depart strongly from this scaling through mechanical impedance, resonance, viscoelasticity, rate-dependent hysteresis and contact dynamics. For PVDF/ZnO, it rises from 3.5 nW at 1 Hz to 352 nW at 100 Hz (Fig. 6). Conduction loss (Eq. 10) acts mainly at low frequency. With $\sigma = 10^{-10}$ S/m plus $\tan \delta = 0.02$, the composite retains 84% of its lossless power at 1 Hz and 98% at 100 Hz. For $\sigma \leq 10^{-11}$ S/m, the loss in power is below 2% at all frequencies. Under matched load, loss mainly lowers the optimum resistance and the stored charge. A semiconducting filler that raises composite conductivity toward 10^{-10} S/m or above would penalise slow, body-motion harvesting more than vibration harvesting.

7.7 Device output

Table 8: Model-predicted device performance under assumed parameters: 2 cm $\times$ 2 cm, 60 μ m film, 50 N, 5 Hz, filler unpoled, nominal nucleation scenario, 10 vol% filler.

Film	d_{33} (pC/N)	ε_r	$d_{33}g_{33}$ (10^{-12} m ² /N)	V_{oc} (V)	R_{opt} (M Ω)	$P_{\max}$ (nW)	$P_{\max, \tan \delta = 0.05}$ (nW)
PVDF/ZnO composite	-24.8	11.7	5.98	1.81	46.3	17.6	16.7
PVDF/BaTiO ₃ composite	-24.8	15.9	4.38	1.32	33.9	12.9	12.3
Unfilled PVDF ($F_{\text{polar}} = 0.50$)	-16.5	12.0	2.55	1.16	44.9	7.5	7.2

Under the assumed mechanical and electrical boundary conditions, particle properties and equal polar-phase content, the PVDF/ZnO composite has a higher model-predicted $d_{33}g_{33}$, V_{oc} and $P_{\max}$ than the PVDF/BaTiO₃ composite (Table 8). The two composites have the same d_{33} . The difference comes entirely from permittivity. This is a controlled model comparison.

Real ZnO and BaTiO₃ particles also differ in surface chemistry, morphology, agglomeration, conductivity, defect chemistry and nucleation efficiency, so the comparison does not establish that one filler is generally superior.

7.8 Parameter sensitivity

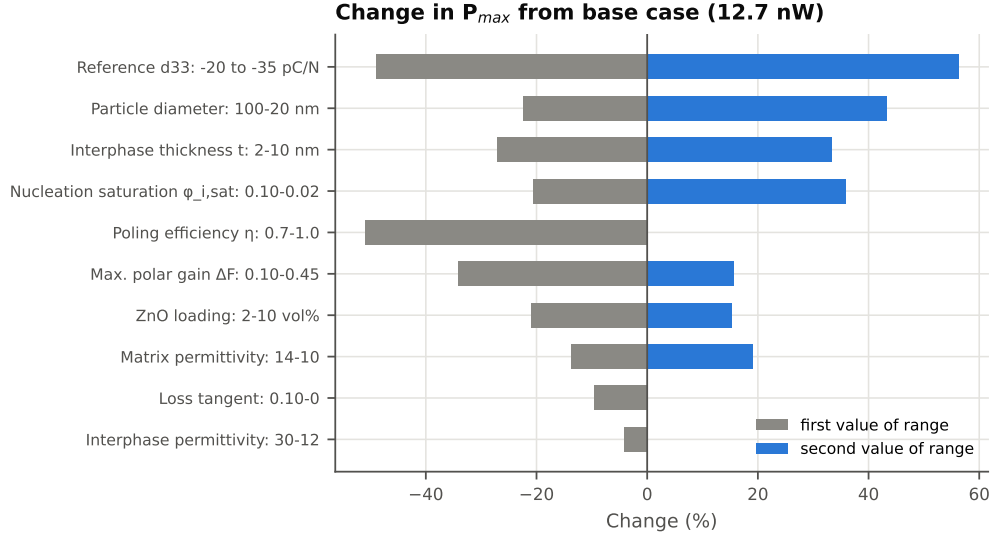


Figure 7: One-at-a-time sensitivity of model-predicted $P_{\max}$ for PVDF/ZnO (50 nm, 5 vol%, interphase-controlled nucleation; base case 12.7 nW). Bars show the change when each parameter is set to the first or second value of its range.

Fig. 7 ranks the parameters by their effect on $P_{\max}$. The largest are the reference matrix coefficient (-49% to $+56\%$) and poling efficiency (down to -51% at $\eta = 0.7$). The next group comprises the parameters that control nucleation: particle diameter (-22% to $+43\%$), nucleation saturation (-21% to $+36\%$), interphase thickness (-27% to $+33\%$) and maximum polar gain (-34% to $+16\%$). Loss tangents up to 0.1 and interphase permittivity (for ZnO) change $P_{\max}$ by 10% or less. The parameters that most need measurement are therefore the matrix d_{33} , the poling efficiency, and the nucleation response of the specific filler. This one-at-a-time analysis does not capture interactions between parameters, which Section 7.9 addresses.

7.9 Global sensitivity and paired comparison

Morris screening varies all parameters over their ranges together (Fig. 8). It agrees with the one-at-a-time ranking at the top: reference $|d_{33}|$ ($\mu^* = 7.3$ nW for ZnO), poling efficiency (5.0 nW) and maximum polar gain (4.1 nW). It also moves the γ -phase parameters into the upper half. For most parameters σ is comparable to μ^* , which indicates strong interactions, as expected for a multiplicative model. One-at-a-time ranges therefore understate the joint uncertainty. With the filler unpoled, the filler permittivity has the smallest effect for both fillers.

A paired Latin-hypercube comparison (2000 samples, the same shared parameters for both fillers) tests whether the ZnO advantage of Table 8 survives this uncertainty. With fillers unpoled, the model predicts a higher $P_{\max}$ for PVDF/ZnO in every sample (ratio 1.07–1.20, 5th–95th percentile). With both fillers poled antiparallel and BaTiO₃ d_f scaled with permittivity, ZnO is again higher in every sample (ratio 1.13–1.40). If BaTiO₃ nanocrystals kept the bulk d_f while having nanocrystal permittivity, ZnO is higher in only 57% of samples (ratio 0.63–1.30). The model-predicted ranking is therefore robust except under that last, physically unlikely, combination.

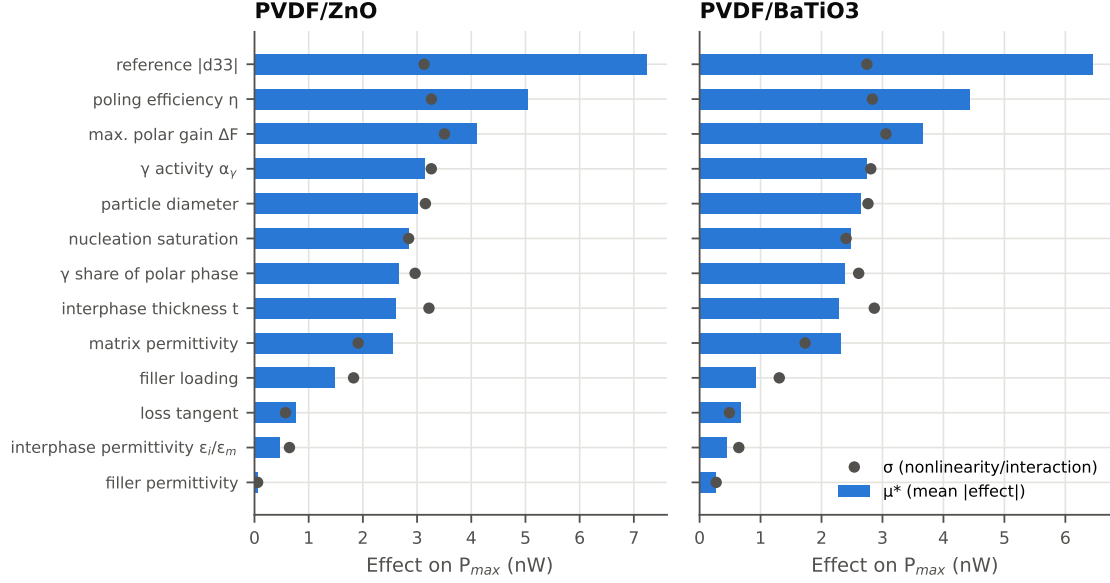


Figure 8: Morris elementary-effects screening of model-predicted $P_{\max}$ (13 parameters, 60 trajectories, 4 levels; filler unpoled). Bars: μ^* , the mean absolute effect over the parameter range. Markers: σ , which indicates nonlinearity or interaction. Parameter ranges are those of Fig. 7, plus: γ share 0–0.5, α_γ 0.25–1, $\varepsilon_i/\varepsilon_m$ 1–2.5, ε_f 8–11 (ZnO) or 80–1700 (BaTiO₃), loading 2–6 vol%.

8 Literature benchmark

Published measurements cannot validate the full chain, but they can check individual assumptions. Table 9 summarises the checks, using only data read from the original articles.

Table 9: External plausibility checks of model assumptions against published measurements.

Quantity	Model	Published	Implication
BaTiO ₃ nanocrystal permittivity	bulk 1700	≈ 80 (≤ 100 nm, paraelectric); 150–300 (≥ 200 nm) [14]	L_E larger; conclusion conditional on d_f scaling (Table 5)
Mixing rule at 20–40 vol%	MG, shaded as extrapolation	MG gives unphysical (negative) inferred particle permittivity at 21–41 vol%; Bruggeman preferred [14]	Supports restricting MG to $\lesssim 15$ vol%
Dense BaTiO ₃ nanocomposite permittivity	–	Lichtenecker with ceramic permittivity overpredicts 28 vol% (24 nm) composites [15]	Bulk filler values overestimate nanocomposite response
Polar crystalline content vs ZnO loading	nominal: $X_c F_{\text{polar}} = 0.25 \rightarrow 0.43$, $\phi_{\text{sat}} = 3$ vol%	0.35 $\rightarrow$ 0.60–0.70 at 0.16–0.64 vol% [16]	Supports a saturating form for this system; saturation much faster; gain largely through X_c

Fig. 9 applies the fitting routine of Section 9 to published FTIR/XRD data for PVDF/ZnO [16]. The saturating law fits both morphologies (r.m.s. error 0.009–0.011). The fitted gains in $X_c F_{\text{polar}}$ are 0.26 (nanoparticles) and 0.34 (nanorods). For the nanorods $\phi_{\text{sat}} = 0.086$ vol%; for the nanoparticles, saturation occurs below the first measured loading of 0.16 vol% and is not resolved. The measured gain comes mostly from crystallinity (41.9% to 65–76%) rather than polar fraction (83% to 91–95%). The model’s fixed X_c therefore understates it, and the nominal ϕ_{sat} of 3 vol% is about 35 times too slow for this system.

These data come from electrospun fibres with in-situ poling and stretching and carbon-coated

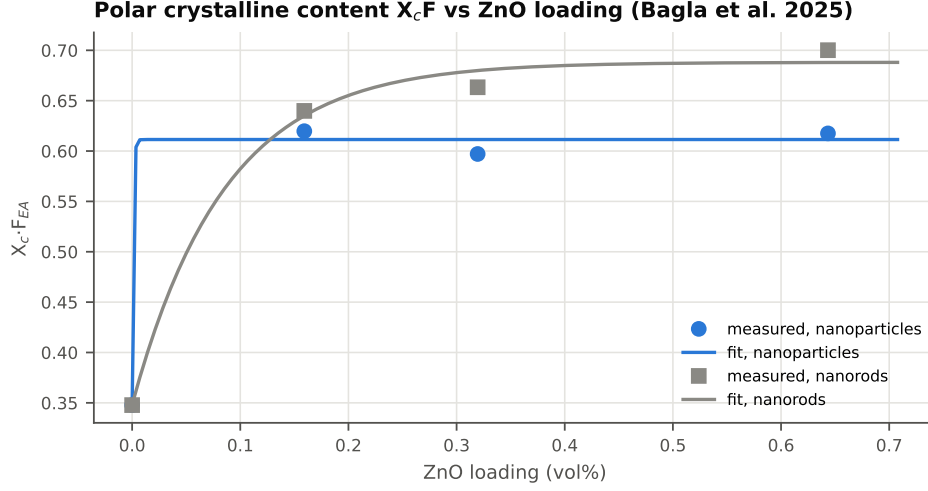


Figure 9: Polar crystalline content $X_c F_{EA}$ of electrospun PVDF with carbon-coated ZnO nanoparticles and nanorods, from Table 1 of Bagla *et al.* [16] (wt% converted to vol% with densities 5.61 and 1.78 g/cm³), with least-squares fits of Eq. (15).

fillers. The polar fraction was computed from the 840 cm⁻¹ band, so it is F_{EA} rather than $F(\beta)$. The values were read from the open preprint of this article, since published in *J. Phys. Chem. C* [16]. They support the applicability of the functional form of Eq. (15) to this specific system but do not fix its parameters for other processing routes.

8.1 Variable crystallinity

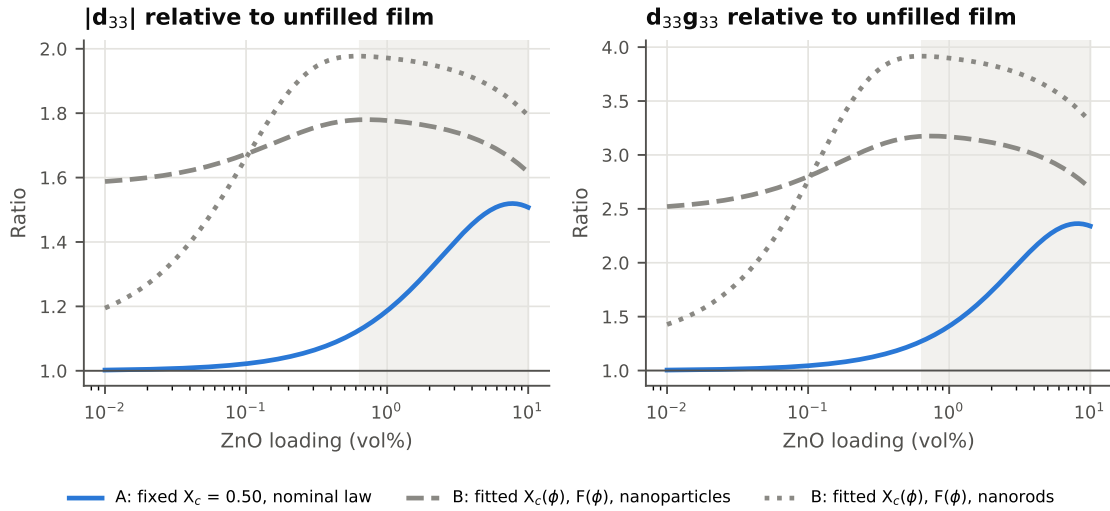


Figure 10: Model-predicted $|d_{33}|$ and $d_{33}g_{33}$ of PVDF/ZnO relative to each case's own unfilled film. Case A: fixed $X_c = 0.50$ with the nominal law. Case B: $X_c(\phi)$ and $F(\phi)$ each fitted to the data of Fig. 9. Shading marks loadings above the largest measured value (0.64 vol%), where case B is extrapolated as saturated.

Because the measured gain comes mainly from crystallinity, we repeated the PVDF/ZnO calculation with X_c allowed to vary (Fig. 10). In case A, X_c is fixed at 0.50 and the nominal law is used. In case B, $X_c(\phi)$ and $F(\phi)$ are each fitted to the published data with Eq. (15); the fits give X_c from 0.42 to 0.66–0.75 and F from 0.83 to 0.92–0.95. At 1 vol%, case B predicts $|d_{33}|$

1.78–1.97 times and $d_{33}g_{33}$ 3.2–3.9 times that of the unfilled film, against 1.19 and 1.41 in case A. At 10 vol% the two cases are closer (1.62–1.79 against 1.51 for $|d_{33}|$), because the nominal law catches up and dilution lowers case B.

Ratios are reported, not absolute values. Applied to the measured saturated values, the linear scaling of Eq. (14) gives a matrix $|d_{33}|$ of 41–46 pC/N, above typical values for poled PVDF. The linear scaling therefore probably overstates the response at high polar content, and case B should be read as an upper estimate of the relative gain. With that caveat, the fixed- X_c results in the rest of the paper are conservative for this system: crystallinity changes can contribute as much as polar-fraction changes, and they appear at much lower loading.

9 Proposed experimental validation protocol

The framework is not yet validated against experiment. The assumed inputs are designed to be replaced by measurements as follows.

1. Measure FTIR (763, 840, 1234, 1275 cm^{-1}) and DSC for at least four loadings, and compute $F(\beta)$, $F(\gamma)$ and X_c with Eq. (1) ff. Report the spectral resolution (e.g. 4 cm^{-1}), the number of co-added scans, the baseline algorithm, and replicate spectra from at least three films per loading. Propagate the replicate scatter into F_{EA} (`phases.electroactive_fraction_`
`uncertainty`) and fit $X_c(\phi)$ as well as $F_{\text{polar}}(\phi)$.
2. Fit Eq. (15) to the measured $F_{\text{polar}}(\phi)$ with the provided least-squares routine (`nucleation.`
`fit_saturating_law`). Use the fitted law, not the scenarios, in Eq. (14).
3. Measure $\varepsilon'(\omega)$ and $\tan \delta(\omega)$ from 0.1 Hz to 1 MHz, and compare with the effective/medium predictions to bound ε_i and t .
4. Measure quasi-static d_{33} on poled samples. Compare it with Eq. (14) using measured X_c and F_{polar} , with η as the only free parameter. Repeat the measurement after short-circuited thermal annealing, for example, to estimate how much of the effective measured d_{33} comes from space charge rather than intrinsic piezoelectricity.
5. Measure V_{oc} and $P(R)$ under controlled force, area and frequency. Include a polarity-reversal test to exclude triboelectric signals.

Agreement in step 4 with a physically reasonable η would support the claim that polar-phase content dominates. Systematic underprediction for piezoelectric fillers would instead point to a larger direct filler contribution or to interfacial space-charge effects.

10 Discussion

Design implications. Within the assumptions of the framework, the following implications follow. They should be tested experimentally.

1. Smaller, well-dispersed particles maximise interphase volume. Agglomerates behave as large particles and introduce voids.
2. The nucleation response of the filler, the β/γ composition it produces and the matrix poling efficiency matter more than the filler’s own d_{33} , provided nanocrystal d_f scales with permittivity. Surface chemistry, which sets surface charge, is therefore a primary variable.
3. A low permittivity mismatch favours both field coupling during poling and g_{33} . High-permittivity piezoceramics need antiparallel or high-field poling to contribute directly.

4. Co-poling a positive- d_{33} filler with PVDF reduces the composite response. The poling sequence and direction should be reported.
5. Harvesters should be compared by $d_{33}g_{33}$ and lossy matched-load power, with force, area, frequency, thickness and load stated.

Limitations. The model is first-order. The dilute-sphere coupling coefficients ignore particle interaction, shape, percolation and imperfect bonding. The matrix term neglects crystal orientation and the amorphous contribution to the sign of d_{33} , and α_γ is unknown. The nucleation laws are phenomenological and assumed. Space-charge electret effects, the size-dependent loss of tetragonality in BaTiO₃ nanoparticles [13], mechanical resonance and nonlinear contact mechanics are not included. The material parameters are representative, not measured.

Open problems. Open problems include separating nucleation, intrinsic filler response, space-charge and triboelectric contributions in measured output. Direct measurement of interphase thickness, permittivity and polarization is also needed, for example by AFM-IR, piezoresponse force microscopy or Kelvin-probe microscopy. Other open problems are the long-term stability of polarization and space charge under humidity, temperature and 10^6 – 10^8 loading cycles, and links from atomistic simulations of surface–chain interactions to continuum electromechanics.

11 Conclusions

We developed a first-order modeling framework that links PVDF phase content, interphase geometry, dielectric mixing, poling configuration and lossy harvester output. Within the parameter range considered, it indicates that changes in matrix polar-phase content can outweigh the directly transmitted response of high-permittivity fillers. Three conditions qualify this. The nucleation law must be measured for each filler system: without a polar-phase gain, the model predicts that fillers lower d_{33} . The nucleated phase must be predominantly β : a large, weakly active γ share removes the gain. And the d_{33} of BaTiO₃ nanocrystals must fall with their permittivity, as the adopted electrostrictive scaling assumes.

Global screening shows that matrix d_{33} , poling efficiency and polar-phase gain dominate the uncertainty, with strong interactions. A paired comparison finds the model-predicted advantage of PVDF/ZnO over PVDF/BaTiO₃ robust except when BaTiO₃ nanocrystals are assumed to retain bulk d_{33} . For the assumed interphase properties, the model predicts an optimum BaTiO₃ size near 17 nm, and conduction loss matters mainly at low frequency. Published data for one electrospun PVDF/ZnO system support a saturating nucleation law but saturate at much lower loading than assumed. Letting crystallinity vary as measured in that system makes the fixed- X_c predictions conservative. The open implementation, the benchmark and the proposed validation protocol allow each assumption to be replaced by measurement.

Code and data availability

The `pvdf_nano` package (version 0.4.0), its 48 unit tests, the script `examples/reproduce_paper.py` that regenerates every figure, table and quoted number, and the generated data files are available at <https://github.com/subrahmanyamtanala-rgb/PVDF-Metal-Oxide-Nanocomposites>. The results were produced with Python 3.11.15, NumPy 2.4.6 and Matplotlib 3.11.2 (`environment-lock.txt`). The full analysis runs in about 5 s on a single CPU core. **[AUTHORS: add license, release tag/commit and archive DOI before submission.]**

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